Module 07: Communication in High School

Learning Objectives

1. Define **Communication** within the context of High School.
2. Analyze how Communication interacts with other components of the Active Inference framework.
3. Apply specific constraints of High School to the formal definition of Communication.

Introduction

This module explores **Communication**. In the **High School** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Communication is a critical component of the 8-part Active Inference spine, bridging the gap between Learning and Planning.

Key Concepts

1. Communication as a Markov Blanket Boundary

How does Communication define the boundary between the agent and the environment?

2. Generative Models of Communication

What parameters involved in Communication must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Communication drive the perception-action loop?

Applications

In High School, we see Communication manifest in: * **Specific Example 1**: The TCP/IP protocol that powers the internet is a communication system built on Active Inference principles: the sender transmits data packets with sequence numbers (predictions of arrival order), the receiver checks whether packets arrive in the expected sequence, and any missing packet (prediction error) triggers a retransmission request -- this error-correction loop ensures reliable communication by minimizing the surprise between sent and received data. * **Specific Example 2**: When two Bluetooth devices pair for the first time, they perform an Active Inference communication handshake: each device broadcasts its identity and capabilities (sharing generative models), they negotiate a shared encryption key (aligning their predictions about how to encode future messages), and once paired, all subsequent communication uses this shared model to minimize the surprise of garbled or intercepted data.

Conclusion

Understanding Communication allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.